1.6.1 Commutative Property

In set language, commutative situations can be seen when we perform operations. For example, we can look into the Union (and Intersection) of sets to find out if the operation is commutative.

Let $A = \{2,3,8,10\}$ and $B = \{1,3,10,13\}$ be two sets.

Then, $A \cup B = \{1,2,3,8,10,13\}$ and

$B \cup A = \{1,2,3,8,10,13\}$

From the above, we see that $A \cup B = B \cup A$.

This is called Commutative property of union of sets.

Now, $A \cap B = \{3,10\}$ and $B \cap A = \{3,10\}$. Then, we see that $A \cap B = B \cap A$.

This is called Commutative property of intersection of sets.

Commutative property: For any two sets A and B

(i) $A \cup B = B \cup A$ (ii) $A \cap B = B \cap A$

Note

For any set A,

- $A \cup A = A$ and $A \cap A = A$ [Idempotent Laws].
- $A \cup \phi = A$ and $A \cap \mathrm{U} = A$ [Identity Laws].

Example 1.19

If $A = \{b,e,f,g\}$ and $B = \{c,e,g,h\}$, then verify the commutative property of (i) union of sets (ii) intersection of sets.

Solution

Given, $A = \{b,e,f,g\}$ and $B = \{c,e,g,h\}$

(i) $A \cup B = \{b,c,e,f,g,h\}$ ... (1)

$B \cup A = \{b,c,e,f,g,h\}$ ... (2)

From (1) and (2) we have $A \cup B = B \cup A$

It is verified that union of sets is commutative.

(ii) $A \cap B = \{e,g\}$ ... (3)

$B \cap A = \{e,g\}$ ... (4)

From (3) and (4) we get, $A \cap B = B \cap A$

It is verified that intersection of sets is commutative.

Thinking Corner

Given, $P = \{l,n,p\}$ and $P \cup Q = \{j,l,m,n,o,p\}$. If P and Q are disjoint sets, then what will be Q and $P \cap Q$?

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